

Gaussian Elimination Method

Solving Linear Systems

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$$Ax = b \quad ; \quad A^{-1} \text{ exist} \\ A \leftrightarrow n \times n \quad \Rightarrow x = A^{-1}b //$$

→ Gauss. eliⁿ.

$$\left. \begin{array}{l} 2x + 3y = 7 \\ x + y = 5 \end{array} \right\} \begin{array}{l} \uparrow \\ \times 2 \end{array} \left. \begin{array}{l} 2x + 3y = 7 \\ 2x + 2y = 10 \end{array} \right\} \begin{array}{l} 2x + 3y = 7 \\ 0x + 1y = -3 \end{array}$$

$$\underline{\underline{y = -3}}$$

Aug. Mat

$$\left(\begin{array}{cc|c} 2 & 3 & 7 \\ 1 & 1 & 5 \end{array} \right) \xrightarrow{R_1} \left(\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & 3 & 7 \end{array} \right)$$

R_2

$$\left(\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 1 & -3 \end{array} \right) \xleftarrow{R_1}$$

$R_2' = R_1 - 2R_2$

Row-Echelon Matrices

REF

Definition (Row-Echelon Matrices)

An $m \times n$ matrix is called a row-echelon matrix if it satisfies the following three conditions:

1. If there are any rows consisting entirely of zeros, they are grouped together at the bottom of the matrix.
2. The first non-zero element in any nonzero row is 1 (called a leading 1).
3. The leading 1 of any row below the first row is to the right of the leading 1 of the row above it.

Example

$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a row-echelon matrix whereas $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix}$ is not.

Reduced Row-Echelon Matrices

Definition

An $m \times n$ matrix is called a reduced row-echelon matrix if it satisfies the following conditions:

1. It is a row-echelon matrix.
2. Any column that contains a leading 1 has zeros everywhere else.

Example (Reduced Row-Echelon Matrices)

$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an RREF. Although, $\begin{pmatrix} 1 & 0.5 \\ 0 & 1 \end{pmatrix}$ is in REF, it's not in RREF.

Motivation

We want to solve a system of three linear equations in three unknowns:

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1, \\ a_{21}x + a_{22}y + a_{23}z = b_2, \\ a_{31}x + a_{32}y + a_{33}z = b_3. \end{cases}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Augmented matrix
(A|b)

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- Gaussian elimination transforms the system into an equivalent but simpler one.
 - The goal is to reduce the system to *upper triangular form*.
- 

Augmented Matrix Form

The system can be written as an augmented matrix:

$$\begin{array}{l} R_1 \longrightarrow \\ R_2 \longrightarrow \\ R_3 \longrightarrow \end{array} \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

Row operations (elementary)
 $\downarrow$
 $\begin{pmatrix} R'_1 \\ R'_2 \\ R'_3 \end{pmatrix}$

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Idea

Use elementary row operations to introduce zeros below the main diagonal.

Elementary Row Operations

The following operations do not change the solution set:

- 1 Interchange two rows: $R_i \leftrightarrow R_j$
- 2 Multiply a row by a nonzero scalar: $R_i \rightarrow cR_i$
- 3 Add a multiple of one row to another:

$$R_i \rightarrow R_i + cR_j$$

Example System in $\mathbb{R}^3$

$$\begin{pmatrix} 0 & 2 & 1 & 4 \\ 2 & 3 & 3 & 10 \\ 3 & 1 & 2 & 7 \end{pmatrix}$$

$$\begin{aligned} 2y + z &= 4 \\ 2x + y + 3z &= 10 \\ 3x + y + 2z &= 7. \end{aligned}$$

Consider the system:

$$\begin{cases} x + 2y + z = 4, \\ 2x + 3y + 3z = 10, \\ 3x + y + 2z = 7. \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 3 & 1 & 2 \end{pmatrix} \text{ \& } b = \begin{pmatrix} 4 \\ 10 \\ 7 \end{pmatrix}$$

$$(A|b) = \begin{pmatrix} 1 & 2 & 1 & 4 \\ 2 & 3 & 3 & 10 \\ 3 & 1 & 2 & 7 \end{pmatrix} \begin{matrix} \longrightarrow R_1 \\ \longrightarrow R_2 \\ \longrightarrow R_3 \end{matrix}$$

$$A = \begin{pmatrix} 1 & 2 & 1 & 4 \\ 2 & 3 & 3 & 10 \\ 3 & 1 & 2 & 7 \end{pmatrix} \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & -1 & 1 & 2 \\ 0 & -5 & -1 & -5 \end{pmatrix}$$

$\det A = c \det A_1$

$$\begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & -6 & -15 \end{pmatrix} \xleftarrow{R_3 + 5R_2} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & -5 & -1 & -5 \end{pmatrix} \xrightarrow{-1R_2}$$

$$\xrightarrow{-1/6 R_3} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 5/2 \end{pmatrix} \Rightarrow$$

$$\begin{aligned} x + 2y + z &= 4 \\ y - z &= -2 \\ z &= 5/2 \end{aligned}$$

Example System in $\mathbb{R}^3$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 3 & 1 & 2 \end{pmatrix} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 3R_1}]{}$$

Consider the system:

$$\begin{cases} x + 2y + z = 4, \\ 2x + 3y + 3z = 10, \\ 3x + y + 2z = 7. \end{cases} \xrightarrow[\substack{-1R_2 \\ -\frac{1}{2}R_3}]{}$$

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 3 & 3 & 10 \\ 3 & 1 & 2 & 7 \end{array} \right] \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 3R_1}]{}$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & -5 & -5 \end{pmatrix} \xleftarrow[-\frac{1}{6}R_3]{\begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & -6 & -15 \end{pmatrix}} \xleftarrow{-1R_2} \begin{pmatrix} 1 & 2 & 1 & 4 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & -6 & -15 \end{pmatrix} \xrightarrow[\substack{R_3 \leftarrow 5R_2}]{}$$

Step 1: Eliminate Below First Pivot

Use the first row as a pivot:

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

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This is an **upper triangular system**.

Back Substitution

From the last row:

$$-6z = -15 \Rightarrow z = \frac{5}{2}$$


Back Substitution

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Second row:

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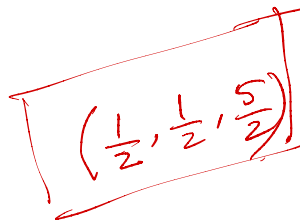
$$-6z = -15 \Rightarrow z = \frac{5}{2}$$

Second row:

$$-y + z = 2 \Rightarrow y = \frac{1}{2}$$

First row:

$$x + 2y + z = 4 \Rightarrow x = \frac{1}{2}$$


$$\left(\frac{1}{2}, \frac{1}{2}, \frac{5}{2}\right)$$

Final Solution

$$(x, y, z) = \left(\frac{1}{2}, \frac{1}{2}, \frac{5}{2} \right)$$

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Geometric Interpretation

The solution corresponds to the unique intersection point of three planes in $\mathbb{R}^3$.

Remarks

The diagram shows an $n \times n$ matrix A with elements a_{ij} . The first column is highlighted in green, and the first row is also highlighted in green. The element a_{21} is circled in red, indicating it is the pivot. A red line is drawn through a_{21} and the rest of the first row, representing row elimination. A blue shaded region covers the submatrix from row 2 to row n and column 2 to column n , with the label A_{11} written inside, representing the $(n-1) \times (n-1)$ submatrix after the first row and column are removed. The matrix is enclosed in large red parentheses, and the word "det" is written to the left.

- Gaussian elimination works for any finite dimension.
- Failure of pivots indicates either:
 - infinitely many solutions, or
 - no solution.
- Computational complexity for 3×3 is negligible, but for $n \times n$ systems it is $\mathcal{O}(n^3)$.

Conclusion

- Gaussian elimination is a fundamental algorithm in linear algebra.
- It provides a systematic way to solve linear systems.
- Gives an idea of LU decomposition and numerical solvers.

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 + cR_1} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} + ca_{11} & a_{22} + ca_{12} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ ca_{11} & ca_{12} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ a_{11} & a_{12} \end{pmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \xrightarrow{R_2 \rightarrow cR_1} \begin{pmatrix} a_{11} & a_{12} \\ ca_{11} & ca_{12} \end{pmatrix}$$

Example:- (Contd) Determine whether the vectors $v_1 = (1, -1, 4)$, $v_2 = (-2, 1, 3)$ and $v_3 = (4, -3, 5)$ span $\mathbb{R}^3$?

Solⁿ: Pick any point/vector in $\mathbb{R}^3$ as $(x_1, x_2, x_3) \in \mathbb{R}^3$.

Qn: Can we find $a, b, c \in \mathbb{R}$ s.t

$$\begin{aligned}(x_1, x_2, x_3) &= av_1 + bv_2 + cv_3 \\ &= a(1, -1, 4) + b(-2, 1, 3) + c(4, -3, 5) \\ &= (a - 2b + 4c, -a + b - 3c, 4a + 3b + 5c)\end{aligned}$$

By equating component-wise, we obtain

$$\left. \begin{aligned}a - 2b + 4c &= x_1 \\ -a + b - 3c &= x_2 \\ 4a + 3b + 5c &= x_3\end{aligned} \right\}$$

Unknowns
are $a, b, c \in \mathbb{R}$

Now apply Gauss elimination method.

Consider the augmented matrix.

$$\begin{pmatrix} 1 & -2 & 4 & x_1 \\ -1 & 1 & -3 & x_2 \\ 4 & 3 & 5 & x_3 \end{pmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 4R_1}]{R_2 \rightarrow R_2 + R_1} \begin{pmatrix} 1 & -2 & 4 & x_1 \\ 0 & -1 & 1 & x_1 + x_2 \\ 0 & 11 & -11 & x_3 - 4x_1 \end{pmatrix} \xrightarrow{R_2 \rightarrow -R_2} \begin{pmatrix} 1 & -2 & 4 & x_1 \\ 0 & 1 & -1 & -x_1 - x_2 \\ 0 & 11 & -11 & x_3 - 4x_1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -2 & 4 & x_1 \\ 0 & 1 & -1 & -x_1 - x_2 \\ 0 & 0 & 0 & 7x_1 + 11x_2 + x_3 \end{pmatrix} \xrightarrow{R_3 - 11R_2} \begin{pmatrix} 1 & -2 & 4 & x_1 \\ 0 & 1 & -1 & -x_1 - x_2 \\ 0 & 0 & 0 & x_3 - 4x_1 + 11(x_1 + x_2) \end{pmatrix}$$